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METHOD AND DEVICE FOR LASER PROCESSING A WORKPIECE

Abstract

A method for laser processing of a workpiece that has a transparent material includes dividing an input laser beam into a plurality of sub-beams by using a beam splitting element, focusing the sub-beams coupled out of the beam splitting element to form multiple focus elements, and applying the focus elements to the material of the workpiece for the laser processing. The focus elements are moved relative to the material in a feed direction. A first sub-quantity of the focus elements is arranged in a first plane. A second sub-quantity of the focus elements is arranged in at least one other plane. The first plane and the at least one other plane are spaced apart in the feed direction. The first plane and the at least one other plane are orientated perpendicular to the feed direction.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/EP2023/076954 (WO 2024/104646 A1), filed on Sep. 28, 2023, and claims benefit to German Patent Application No. DE 10 2022 130 022.6, filed on Nov. 14, 2022. The aforementioned applications are hereby incorporated by reference herein.

FIELD
[0002] Embodiments of the present invention relate to a method for laser processing of a workpiece having a transparent material. Embodiments of the present invention also relate to a device for laser processing of a workpiece.
BACKGROUND
[0003] WO 2022/167254 A1 and WO 2022/167257 A1 each disclose methods and devices for laser processing of a transparent workpiece, wherein the workpiece is applied with a multitude of focus elements for the laser processing.
[0004] DE 10 2014 116 958 A1 discloses a diffractive optical beam forming element for applying a phase profile to a laser beam provided for laser processing of a material substantially transparent to the laser beam using a phase mask, which is formed for applying a plurality of beam-forming phase profiles to the laser beam incident on the phase mask, and at least one of the plurality of beam-forming phase profiles is associated with a virtual optical image, which can be imaged in at least one elongated focus zone to form a modification in the material to be processed.
[0005] A method for severing a transparent material using an elongated focus zone of a laser beam is known from EP 3 597 353 A1.
[0006] A method for severing and in particular beveling a transparent material is known from JP 2020 004 889 A, wherein a plurality of focal points for laser processing of the material are generated using a spatial light modulator.
[0007] Methods for forming a beveled edge region on a transparent material using a laser beam are known from US 2020/0147729 A1 and US 2020/0361037 A1.
[0008] A method for severing a transparent material using multiple parallel non-diffracting laser beams is known from WO 2016/089799 A1.
SUMMARY
[0009] Embodiments of the present invention provide a method for laser processing of a workpiece that has a transparent material. The method includes dividing an input laser beam into a plurality of sub-beams by using a beam splitting element, focusing the sub-beams coupled out of the beam splitting element to form multiple focus elements, and applying the focus elements to the material of the workpiece for the laser processing. The focus elements are moved relative to the material in a feed direction. A first sub-quantity of the focus elements is arranged in a first plane. A second sub-quantity of the focus elements is arranged in at least one other plane. The first plane and the at least one other plane are spaced apart in the feed direction. The first plane and the at least one other plane are orientated perpendicular to the feed direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

[0011] FIG. **1** shows a schematic illustration of one exemplary embodiment of a device for laser processing of a workpiece;

[0012] FIG. **2** shows a schematic cross-sectional view of a portion of a material of the workpiece, in which multiple focus elements are applied to the material for laser processing according to some embodiments;

[0013] FIG. **3** shows a schematic cross-sectional view of a portion of the workpiece in which material modifications, which are accompanied by crack formation in the material, were produced by applying focus elements to the workpiece, according to some embodiments;

[0014] FIG. **4** shows an arrangement of focus elements designed for laser processing of the workpiece, wherein a projection of the focus elements is shown in a projection plane orientated perpendicular to the feed direction, according to some embodiments;

[0015] FIG. **5***a* shows an arrangement of focus elements designed for laser processing of the workpiece in a cross-section orientated parallel to the feed direction and to the thickness direction of the workpiece, according to some embodiments;

[0016] FIG. **5***b* shows the focus elements according to FIG. **5***a* in a cross-section orientated parallel to the feed direction and perpendicular to the thickness direction of the workpiece, according to some embodiments;

[0017] FIG. **6** shows the focus elements according to FIGS. **5***a* and **5***b* in a perspective view, according to some embodiments;

[0018] FIG. **7***a* shows a cross-sectional view of a simulated intensity distribution of focus elements for laser processing of the workpiece, wherein mutually adjacent focus elements are each spaced apart at a distance of approximately 17.5 μm, according to some embodiments;

[0019] FIG. **7***b* shows a cross-sectional view of a simulated intensity distribution of focus elements for laser processing of the workpiece, wherein mutually adjacent focus elements are each spaced apart at a distance of approximately 8.0 μm, according to some embodiments;

[0020] FIG. **8***a* shows a schematic perspective view of a workpiece having material modifications produced thereon, which extend along a processing line and/or processing surface, according to some embodiments; and

[0021] FIG. **8***b* shows a schematic perspective view of two workpiece segments, which are formed by severing the workpiece according to FIG. **8***a* along the processing line and/or processing surface, according to some embodiments;

DETAILED DESCRIPTION

[0022] Embodiments of the present invention provide a method and a device by means of which material modifications can be formed in the material of the workpiece, which enable a severing of the material with an improved quality and in particular a reduced roughness at the severing surface.

[0023] According to some embodiments, in a method for laser processing of a workpiece having a transparent material, an input laser beam is divided into a plurality of sub-beams by means of a beam splitting element, and sub-beams coupled out of the beam splitting element are focused, multiple focus elements are formed by focusing the sub-beams, focus elements are applied to the material of the workpiece for the laser processing, and the focus elements are moved relative to the material in a feed direction. A sub-quantity of the formed focus elements is arranged in a first plane and a sub-quantity of the formed focus elements is arranged in at least one further plane, wherein the first plane and the at least one further plane are spaced apart parallel to the feed direction and

wherein the first plane and the at least one further plane are orientated perpendicular to the feed direction.
[0024] During the laser processing of the workpiece by means of the method according to embodiments of the invention, material modifications are produced in the material of the workpiece which in particular enable severing of the material.
[0025] By arranging focus elements both in the first plane and in at least one further plane, an effective distance of the focus elements orientated perpendicular to the feed direction can be reduced. This allows adjacent material modifications to be produced in the material of the workpiece with a particularly small distance between them, which in turn increases the density of the material modifications formed in the material. This improves the severability of the material. In addition, the quality of a severing surface resulting from the severing of the material can be increased, and the severing surface can in particular be designed with a reduced roughness and/or increased smoothness. This results in increased edge stability of the workpiece material at the severing surface after severing has taken place.
[0026] If an actual distance of the mutually adjacent focus elements becomes too small, undesired interference effects between adjacent focus elements can arise from this, which can have the consequence, for example, of beat effects in the intensity of the focus elements. This can make it difficult to control the production of the material modifications and in particular make it difficult to produce identical material modifications. In the solution according to embodiments of the invention, the focus elements for producing the material modifications are arranged ‘offset’ in the feed direction, which results in a significantly reduced effective distance when the actual distance between the focus elements is sufficiently large.
[0027] The distance or actual distance between focus elements is the actual distance between these focus elements in three-dimensional space. The effective distance between the focus elements is the distance between perpendicular projections of the focus elements in a projection plane orientated perpendicular to the feed direction.
[0028] A spatial position and/or arrangement of a specific focus element is to be understood in particular as that of its center and/or center of gravity within the material. In particular, the distance and the effective distance are related to the corresponding centers of the focus elements in the material, i.e. they are center distances.
[0029] By relative movement of the focus elements arranged in the first plane and the at least one further plane with respect to the material in the feed direction, material modifications are produced at least in portions of the material which are spaced apart by the effective distance.
[0030] With the method according to embodiments of the invention, the material modifications can be produced in particular in such a way that adjacent material modifications overlap in the material. This results in a crack connection between the adjacent material modifications, which enables the material to be severed particularly well by etching or applying heat.
[0031] By applying the focus elements to the material of the workpiece at a specific point in time, material modifications are produced which are located in the material at positions where the focus elements are positioned in the material at that point in time.
[0032] In particular, all focus elements associated with the first plane and/or the at least one further plane are present simultaneously.
[0033] For example, two or more planes orientated perpendicular to the feed direction can be provided, in each of which a sub-quantity of the formed focus elements is arranged. The intended planes are orientated parallel to each other and/or spaced apart from each other in the feed direction.
[0034] In particular, it is possible for a distance between the first plane and the at least one further plane to at least 2 μm and/or at most 200 μm. For example, the distance is approximately 10 μm.
[0035] In particular, it is possible for the formed focus elements to be positioned such that, when the focus elements are viewed in a projection plane orientated perpendicular to the feed direction,

at least one sub-quantity of focus elements associated with the first plane and focus elements associated with the at least one further plane are arranged at different spatial positions in the projection plane. This makes it possible to reduce the effective distance between adjacent focus elements, such that the distance between formed adjacent material modifications is reduced accordingly.
[0036] In particular, it is possible for the formed focus elements to be positioned such that, when viewing the focus elements in a projection plane orientated perpendicular to the feed direction, at least a sub-quantity of focus elements associated with the first projection plane and focus elements associated with the at least one further projection plane are positioned in the projection plane along a processing line, wherein, by relative movement of the focus elements in relation to the material in the feed direction, material modifications are produced in the material which are arranged along the processing line. In particular, an edge geometry and/or a cross-sectional geometry of a severing surface arising due to severing of the material at the material modifications may be defined by means of the processing line.
[0037] For example, the at least one processing line has a total length of between 10 μm and 10000 μm and in particular between 100 μm and 1000 μm and in particular between 400 μm and 600 μm. Workpieces having a thickness in the stated range can thus be processed and in particular severed.
[0038] The material of the workpiece has, for example, a thickness of between 10 μm and 10000 μm and preferably between 100 μm and 1000 μm, for example approximately 500 μm.
[0039] In particular, it is possible for the processing line to be formed spatially continuously over a thickness of the material of the workpiece and/or over a thickness of a workpiece segment to be detached from the workpiece.
[0040] The processing line is not necessarily of spatially contiguous design, but rather may have different, spatially separate portions. In particular, the processing line can have interruptions, in which no focus elements are arranged.
[0041] In particular, material modifications along a processing surface corresponding to the processing line are produced by the relative movement of the focus elements in the material. The material modifications are then arranged along the processing line, particularly when viewed in a cross-section, orientated perpendicular to the feed direction, through the processing surface. After the workpiece has been severed at the material modifications arranged along the processing surface, a shape and/or cross-sectional shape of the severing surface formed during the severing then corresponds in particular to a shape and/or cross-sectional shape of the (previous) processing surface.
[0042] It may be advantageous if adjacent focus elements arranged along the processing line have an effective distance of at least 2 μm and/or at most 200 μm in the projection plane. This allows the severing surface to be produced with particularly high quality and/or reduced roughness.
[0043] The focus elements arranged along the processing line and/or viewed in the projection plane are understood to mean the focus elements which are associated with the first plane and the at least one further plane.
[0044] In particular, the effective distance of the focus elements along the processing line corresponds at least approximately to a distance of material modifications which are formed on the processing surface corresponding to the processing line and are positioned in a cross-section, orientated perpendicular to the feed direction, through the processing surface.
[0045] In particular, it is possible for an effective distance of adjacent focus elements arranged along the processing line in the projection plane and/or an intensity of the focus elements to be selected such that, by applying the focus elements to the material and relative movement of the focus elements in relation to the material in the feed direction, material modifications are produced in the material which enable a severing of the material.
[0046] In particular, it is possible for at least a sub-quantity of adjacent focus elements arranged along the processing line to at least approximately spaced apart in the projection plane with the

same effective distance from one another.

[0047] It may be advantageous that at least a sub-quantity of adjacent focus elements arranged along the processing line is spaced apart at an effective distance, the effective distance having an effective distance component different to zero which is orientated parallel to a thickness direction of the workpiece, and/or having a further effective distance component different to zero which is orientated perpendicular to the thickness direction of the workpiece. Thus, for example, a perpendicular cut can be produced on the workpiece or the workpiece can be chamfered at a specific angle.

[0048] The thickness direction of the workpiece is to be understood in particular to mean a direction which is orientated transverse and in particular perpendicular to an outer side of the workpiece through which the focus elements and/or a laser beam for forming the focus elements are coupled into the material.

[0049] The thickness direction is orientated in particular transversely or perpendicularly to a beam propagation direction of a laser beam from which the focus elements are formed.

[0050] In particular, it is possible for at least a sub-quantity of mutually adjacent focus elements arranged along the processing line to be spaced apart at an effective distance, the effective distance having an effective distance component different to zero which is orientated parallel to a beam propagation direction of a laser beam from which the focus elements are formed, and/or for at least a sub-quantity of mutually adjacent focus elements arranged along the processing line to be spaced apart at an effective distance, the effective distance having an effective distance component different to zero which is orientated perpendicular to a beam propagation direction of a laser beam from which the focus elements are formed.

[0051] For the same reason, it can be advantageous if an angle of attack between the processing line and an outer side of the workpiece, through which the focus elements for laser processing of are coupled into the material of the workpiece, is at least 1° and/or at most 90° and in particular at most 89°, at least in some portions. Depending on the selection of the angle of attack, for example, a perpendicular cut may thus be executed on the workpiece or the workpiece may be chamfered at a specific angle.

[0052] The processing line having a specific angle of attack or angle of attack range at least in places is to be understood in particular to mean that the processing line has at least one portion with this angle of attack or angle of attack range.

[0053] In particular, the angle of attack can be at least 10° and/or at most 80°, preferably at least 30° and/or at most 60°, particularly preferably at least 40° and/or at most 50°.

[0054] In particular, it is possible for the angle of attack of the processing line to be constant at least in places, and/or for the processing line to have multiple portions with different angles of attack.

[0055] In particular, it is possible for the processing line to be a straight line at least in portions, and/or that the processing line is a curve at least in portions.

[0056] By configuring the processing line as a curve, for example, rounded segments can be detached from the workpiece. For example, rounded edges can thus be created.

[0057] If the processing line is configured as a curve, the processing line is, for example, associated with a specific angle of attack range, which the processing line has with respect to an outer side of the workpiece.

[0058] It can be advantageous if the formed focus elements are positioned such that, when the focus elements are viewed in a projection plane orientated perpendicular to the feed direction, gaps are formed between mutually adjacent focus elements associated with the first plane, there being focus elements associated with the at least one further plane which are arranged in the gaps. This makes it possible to reduce the effective distance between adjacent focus elements and/or increase the density of material modifications produced in the material.

[0059] In particular, it is possible for a distance between mutually adjacent focus elements

associated with the first plane to be at least 3 μm and/or at most 70 μm and in particular at least 5 μm and/or at most 10 μm.
[0060] In particular, it is possible for a distance between mutually adjacent focus elements associated with the at least one further plane to be at least 3 μm and/or at most 70 μm and in particular at least 5 μm and/or at most 10 μm.
[0061] In particular, it is possible for a dividing of the input laser beam by means of the beam splitting element to take place by phase imprinting on a beam cross-section of the input laser beam or to comprise a phase imprinting on a beam cross-section of the input laser beam. The focus elements can thus be formed, for example, as copies of one another. In particular, the focus elements can thereby be introduced in a technically simple manner at different positions and/or with different spacings into the material of the workpiece.
[0062] It is possible for the dividing of the input laser beam to take place exclusively by phase imprinting on the beam cross-section of the input laser beam.
[0063] In particular, the phase imprinting takes place in the transverse direction of the input laser beam. The transverse direction lies in a plane orientated perpendicular to the beam propagation direction of the input laser beam.
[0064] Alternatively or additionally, it is possible for a dividing of the input laser beam by means of the beam splitting element to take place by polarization beam splitting or to comprise polarization beam splitting. Then, mutually adjacent focus elements can, for example, each be formed with different polarization states. This makes it possible to prevent interference between mutually adjacent focus elements, which means that they can be arranged at a particularly small distance from one another.
[0065] It is possible in principle for the dividing of the input laser beam to take place both by means of phase imprinting and by means of polarization beam splitting.
[0066] In particular, the input laser beam and/or a laser beam from which the focus elements are formed is a pulsed laser beam and in particular an ultrashort pulse laser beam. By applying the focus elements to the material, in particular laser pulses and in particular ultrashort laser pulses are thereby introduced into the material.
[0067] The material modifications introduced into transparent materials by ultrashort laser pulses are subdivided into three different classes; see K. Itoh et al. “Ultrafast Processes for Bulk Modification of Transparent Materials” MRS Bulletin, vol. 31, p. 620 (2006): type I is an isotropic refractive index change; type II is a birefringent refractive index change; and type III is a void or cavity. The type of material modification produced by a particular focus element depends on laser parameters of the laser beam from which the corresponding focus element is formed, such as the pulse duration, the wavelength, the pulse energy and the repetition frequency of the laser beam. Furthermore, the type of modification depends on the properties of the material, such as the electronic structure and the thermal coefficient of expansion of the material, as well as the numerical aperture (NA) used when focusing the laser beam into the corresponding focus element.
[0068] Type I isotropic refractive index changes are attributed to spatially restricted melting by the laser pulses and fast resolidification of the transparent material. For example, quartz glass has a higher material density and refractive index if the quartz glass is cooled down quickly from a higher temperature. Thus, if the material in the volume detected by the focus element melts and then cools rapidly, the quartz glass has a higher refractive index in the regions of material modification than in the unmodified regions.
[0069] Type II birefringent refractive index changes may arise, for example, due to interference between an ultrashort laser pulse and the electric field of the plasma generated by the laser pulses. This interference leads to periodic modulations in the electron plasma density, which upon solidification leads to a birefringent property, that is to say direction-dependent refractive indices, of the transparent material. Type II modification is, for example, also accompanied by the formation of nanogratings.

[0070] In particular, the voids (cavities) of the type III modifications can be produced using a high laser pulse energy. Formation of the voids is in this case attributed to an explosive expansion of highly excited, vaporized material from the focus volume into the surrounding material. This process is also referred to as microexplosion. Since this expansion occurs within the mass of the material, the microexplosion leaves behind a less dense or hollow core (the void), or a microscopic defect in the sub-micrometer range or in the atomic range, which void or defect is surrounded by a compacted material envelope. Compaction at the shock front of the microexplosion creates stresses in the transparent material, which regularly lead to spontaneous crack formation or promote crack formation.

[0071] Therefore, when the introduction of a type III modification is concerned, then a less dense or hollow core or a defect is present in any case. By way of example, in the case of a type III modification in sapphire, a region of lower density rather than a void is produced by the microexplosion.

[0072] In particular, the formation of voids can also be accompanied by type I and type II modifications. By way of example, type I and type II modifications can arise in the less stressed areas around the introduced laser pulses. The formation of type I and type II modifications cannot be fully prevented or avoided when introducing type III modifications. It is therefore unlikely that “pure” type III modifications are to be found.

[0073] It can be advantageous if material modifications are produced in the material by applying the focus elements to the material of the workpiece, the material modifications being accompanied by crack formation in the material, and/or the material modifications being type III material modifications. In particular, the material can be severed by means of these material modifications.

[0074] It can be favorable if material modifications are formed in the material by applying the focus elements to the material of the workpiece, the material modifications being accompanied by a change of a refractive index of the material, and/or the material modifications being type I material modifications and/or type II material modifications. In particular, the material can be severed by means of these material modifications.

[0075] In particular, it is possible for the material of the workpiece to be severable or severed after laser processing, with it being possible in particular for the material to be severable or severed at a processing surface where material modifications were produced by means of the laser processing.

[0076] In particular, it is possible for the material of the workpiece to be severable or severed by applying heat and/or a mechanical tension and/or by etching by means of at least one wet-chemical solution. For example, the etching takes place in an ultrasound-assisted etching bath. The application of heat can be achieved, for example, by means of a CO.sub.2 laser.

[0077] According to some embodiments, a device for laser processing of a workpiece having a transparent material includes a beam splitting element for dividing an input laser beam into a plurality of sub-beams, a focusing optical unit for focusing sub-beams coupled out of the beam splitting element, multiple focus elements for laser processing of the workpiece being formed by focusing the sub-beams, and a feed apparatus for carrying out a relative movement of the focus elements relative to the material of the workpiece in a feed direction. The beam splitting element and the focusing optical unit are configured to arrange the focus elements in such a way that a sub-quantity of the formed focus elements are arranged in a first plane and a sub-quantity of the formed focus elements are arranged in at least one other plane, the first plane and the at least one other plane being spaced apart in parallel with the feed direction, and the first plane and the at least one other plane being orientated perpendicular to the feed direction.

[0078] In particular, the device according to embodiments of the invention has one or more further features and/or advantages of the method according to embodiments of the invention. Advantageous embodiments of the device according to embodiments of the invention have already been explained in connection with the method according to embodiments of the invention.

[0079] In particular, the method according to embodiments of the invention can be carried out by

means of the device according to embodiments of the invention, or the method according to embodiments of the invention is carried out by means of the device according to embodiments of the invention.
[0080] It can be advantageous if the beam splitting element is formed as a 3D beam splitting element or comprises a 3D beam splitting element. It is then possible for the dividing of the input laser beam to take place by phase imprinting on a beam cross-section of the input laser beam and in particular exclusively by phase imprinting on the beam cross-section of the input laser beam.
[0081] It can be favorable if the beam splitting element is formed as a polarization beam splitting element or comprises a polarization beam splitting element.
[0082] For example, the beam splitting element comprises multiple components and/or functionalities. It is possible for the beam splitting element to comprise both a 3D beam splitting element and a polarization beam splitting element.
[0083] In particular, the device comprises a laser beam source for providing the input laser beam, the input laser beam being in particular a pulsed laser beam and/or an ultrashort pulse laser beam.
[0084] In particular, a transparent material is to be understood to mean a material through which at least 70% and in particular at least 80% and in particular at least 90% of a laser energy of the input laser beam and/or of a laser energy of a laser beam from which the focus elements are formed is transmitted.
[0085] In particular, a focus element is to be understood as a radiation region having a specific spatial extension and intensity distribution. To determine spatial dimensions of a specific focus element, such as, for example, a diameter of the focus element, only intensity values of the intensity distribution above a specific intensity threshold are considered. In this regard, the intensity threshold is selected, for example, such that values below this intensity threshold have such a low intensity that they are no longer relevant for interaction with the material for the purpose of producing material modifications. For example, the intensity threshold is 50% of a global intensity maximum of the focus element.
[0086] In particular, a specific focus element is associated with a spatial region of interaction, in which the focus element interacts with the material of the workpiece when it is introduced into this material.
[0087] In particular, the focus elements introduced into the material interact with the material by nonlinear absorption. In particular, material modifications are produced in the material due to nonlinear absorption by means of the focus elements.
[0088] In particular, it is possible for the focus elements according to the preceding definition to have a maximum spatial extension of at least 0.5 μm and/or at most 30 μm, preferably at least 2 μm and/or at most 10 μm. In particular, a maximum spatial extension of a region of interaction, associated with a given focus element, with the material of the workpiece is at least 0.5 μm and/or at most 30 μm, and preferably at least 2 μm and/or at most 10 μm.
[0089] The maximum spatial extension of a given focus element is to be understood in particular to mean the greatest spatial extension of the focus element in any spatial direction.
[0090] In particular, a maximum spatial extension of the focus elements is less than 20% and preferably less than 10% and particularly preferably less than 5% of a thickness of the material.
[0091] In particular, the focus elements have a diffracting beam profile. In particular, the focus elements are designed to be diffraction-limited. For example, a specific focus element has a Gaussian shape and/or a Gaussian intensity profile.
[0092] In particular, the input laser beam and/or a laser beam from which the focus elements are formed has a diffracting beam profile and/or a Gaussian beam profile.
[0093] For example, a wavelength of the input laser beam and/or of the laser beam from which the focus elements are formed is at least 300 nm and/or at most 1500 nm. For example, the wavelength is 515 nm or 1030 nm.
[0094] In particular, the input laser beam and/or the laser beam from which the focus elements are

formed has an average power of at least 1 W to 1 kW. For example, the laser beam comprises pulses having a pulse energy of at least 10 μJ and/or at most 50 mJ. It is possible for the laser beam to comprise individual pulses or bursts, with the bursts having 2 to 20 subpulses and in particular a time interval of approximately 20 ns.

[0095] The statement “at least a sub-quantity of the focus elements” can mean either a sub-quantity of the focus elements or a total amount of the focus elements, i.e. all focus elements.

[0096] In particular, the statements “at least approximately” or “approximately” should be understood to mean in general a deviation of at most 10%. Unless stated otherwise, the statements “at least approximately” or “approximately” are to be understood to mean in particular that an actual value and/or distance and/or angle deviates by no more than 10% from an ideal value and/or distance and/or angle.

[0097] Elements that are the same or have equivalent functions are provided with the same reference signs in all of the figures.

[0098] One exemplary embodiment of a device for laser processing of a workpiece is shown in FIG. **1** and denoted by **100** therein. Localized material modifications, such as defects in the sub-micrometer range or atomic range, which result in material weakening, may be produced in a material **102** of the workpiece **104** using the device **100**. At these material modifications, the workpiece **104** can be severed, wherein a workpiece segment can, for example, be detached from the workpiece **104**.

[0099] In particular, material modifications can be introduced into the material **102** at an angle of attack using the device **100**, such that an edge region of the workpiece **104** can be chamfered or beveled by detaching a workpiece segment from the workpiece **104**.

[0100] The device comprises a beam splitting element **106**, into which an, in particular collimated, input laser beam **108** is coupled. This input laser beam **108** is provided, for example, by a laser beam source **110**. In particular, the input laser beam **108** is a pulsed laser beam and/or an ultrashort pulse laser beam.

[0101] It is possible for the laser beam source **110** to comprise a hollow-core fiber (not shown) from which a laser beam formed by means of the laser beam source emerges. This laser beam is then collimated, for example, by means of a collimation optical unit (not shown) of the laser beam source **110** to form the collimated input laser beam **108**.

[0102] In particular, the input laser beam **108** is to be understood to mean a beam bundle comprising a plurality of beams extending in particular in parallel. The input laser beam **108** in particular has a transverse beam cross-section **112** and/or a transverse beam expansion with which the input laser beam **108** impinges on the beam splitting element **106**. The input laser beam **108** impinging on the beam splitting element **106** in particular has at least approximately planar wavefronts **114**.

[0103] The input laser beam **108** is divided by the beam splitting element **106** into a plurality of sub-beams **116** and/or sub-beam bundles. In the example shown in FIG. **1**, two different sub-beams **116***a* and **116***b* are indicated.

[0104] In particular, the beam splitting element **106** takes the form of a far-field beam forming element. The sub-beams **116** or sub-beam bundles coupled out of the beam splitting element **106** in particular have a divergent beam profile and/or propagate in the manner of spherical waves.

[0105] To focus the sub-beams **116** coupled out of the beam splitting element **106**, the device **100** comprises a focusing optical unit **118**, into which the sub-beams **116** are coupled. The focusing optical unit **118** has one or more lens elements, for example. For example, the focusing optical unit **118** is formed as a microscope objective.

[0106] For example, the focusing optical unit **118** has a focal length between 5 mm and 50 mm.

[0107] The beam splitting element **106** is in particular at least approximately arranged in a rear focal plane of the focusing optical unit **118**.

[0108] In particular, different sub-beams **116** impinge on the focusing optical unit **118** with a position offset and/or angular offset. These sub-beams **116** are focused by means of the focusing optical unit **118** so that multiple focus elements **120** are formed, which are each arranged at different spatial positions. It is possible in principle for mutually adjacent focus elements to spatially overlap in portions.

[0109] For example, one or more sub-beams **116** and/or sub-beam bundles are each associated with a given focus element **120**. For example, a focus element **120** is formed by focusing one or more sub-beams **116** and/or sub-beam bundles.

[0110] A focus element **120** is to be understood in particular as a focused radiation region, such as a focus spot and/or a focus point. In particular, the focus elements **120** each have a specific geometric shape and/or a specific intensity profile, and the geometric shape is to be understood, for example, as a spatial shape and/or a spatial extension of the focus element **120**.

[0111] The geometric shape and/or the intensity profile of a specific focus element **120** is referred to hereinafter as the focus distribution **121** of the focus element **120**. The focus distribution **121** is a property of the focus elements **120** and describes their shape and/or intensity profile. In particular, multiple focus elements **120** or all focus elements **120** formed have the same focus distribution.

[0112] The focus distribution of the focus elements **120** formed is defined by the input laser beam **108**, the dividing of which using the beam splitting element **106** forms the focus elements **120**. If the input laser beam **108** were focused before being coupled into the beam splitting element **106**, a single focus element would thus be formed having the focus distribution associated with the input laser beam **108**.

[0113] For example, the input laser beam **108**, if it is provided, for example, by means of the laser beam source **110**, has a Gaussian beam profile. By focusing the input laser beam **108**, a focus element would be formed in this case, which has a focus distribution with a Gaussian shape and/or Gaussian intensity profile.

[0114] Alternatively thereto, for example, it is possible for a Bessel-like beam profile to be associated with the input laser beam **108**, so that, by focusing the input laser beam **108**, a focus element would be formed which has a focus distribution having Bessel-like shape and/or Bessel-like intensity profile.

[0115] The focus distribution of the input laser beam **108** is associated with the sub-beams **116** and/or sub-beam bundles formed by dividing of the input laser beam **108** using the beam splitting element **106** such that by focusing the sub-beams **116**, the focus elements **120** are formed with this focus distribution and/or with a focus distribution based on this focus distribution.

[0116] In the example shown in FIG. **1**, the input laser beam **108** has a Gaussian beam profile, i.e., a focus distribution with a Gaussian shape and/or Gaussian intensity profile is associated with the input laser beam **108**. The focus elements **120** then each, for example, have the focus distribution **121** having this Gaussian shape and/or this Gaussian intensity profile or having a shape and/or intensity profile based on this Gaussian shape and/or this Gaussian intensity profile (cf. also FIGS. **5***a* and **5***b*).

[0117] If, for example, a Bessel-like beam profile is associated with the input laser beam **108**, the focus elements **120** formed for laser processing of the workpiece **104** each have a focus distribution **121** having this Bessel-like beam profile or having a beam profile based on this Bessel-like profile. The focus elements **120** can thus each be formed, for example, with a focus distribution that has an elongate shape and/or an elongate intensity profile.

[0118] It is possible for the device **100** to have a beam forming apparatus **122** for beam forming of the input laser beam **108** (indicated in FIG. **1**). For example, this beam forming apparatus **122** is arranged upstream of the beam splitting element **106** and/or between the laser beam source **110** and the beam splitting element **106** with respect to a beam propagation direction **124** of the input laser beam **108**.

[0119] A beam propagation direction is to be understood in particular to mean a main beam propagation direction and/or an average propagation direction of a laser beam and/or beam bundle.

The beam propagation direction corresponds in particular to a direction of a Poynting vector associated with the laser beam or beam bundle.

[0120] By means of the beam forming apparatus **122**, a specific beam profile can be associated with the input laser beam **108**, which defines the focus distribution **121** of the focus elements **120**.

[0121] The beam forming apparatus **122** can be configured, for example, to form a laser beam with a quasi-non-diffracting and/or Bessel-like beam profile from a laser beam with a Gaussian beam profile. For this purpose, the beam forming apparatus **122** is or comprises, for example, an axicon element.

[0122] The input laser beam **108** coupled into the beam splitting element **106** then has the quasi-non-diffracting and/or Bessel-like beam profile. The focus elements **120** then accordingly also have this quasi-non-diffracting and/or Bessel-like beam profile or a beam profile based on this beam profile.

[0123] With regard to the definition and implementation of quasi-non-diffracting and/or Bessel-like beams, reference is made to the book: “Structured Light Fields: Applications in Optical Trapping, Manipulation and Organisation”, M. Wördemann, Springer Science & Business Media (2012), ISBN 978-3-642-29322-1 and also to the scientific publications “Bessel-like optical beams with arbitrary trajectories” by I Chremmos et al., Optics Letters, vol. 37, no. 23, 1 Dec. 2012 and “Generalized axicon-based generation of nondiffracting beams” by K. Chen et al., arXiv:1911.03103v1 [physics.optics], Nov. 8, 2019.

[0124] The focus elements **120** are in particular each designed to be identical to one another and/or are each designed as copies of one another by beam splitting using the beam splitting element **106**.

[0125] A specific local position x.sub.0, y.sub.0, z.sub.0 is associated with each of the focus elements **120** formed, at which position a focus element **120** is arranged with respect to the material **102** of the workpiece **104** (FIG. **2**). For example, the spatial position of a focus element **120** is to be understood to mean the position of its spatial center and/or center of gravity.

[0126] Furthermore, a specific intensity I is in particular associated with each of the focus elements **120** formed. The spatial position x.sub.0, y.sub.0, z.sub.0 and in particular also the intensity I of the focus elements **120** can be adjusted using the beam splitting element **106**.

[0127] In particular, multiple or all focus elements **120** formed for laser processing of the workpiece **104** have the same intensity I. However, it is also possible for multiple focus elements **120** formed to have different intensities I.

[0128] In particular, by means of the beam splitting element **106**, a distance d and/or a spatial offset between adjacent focus elements **120** can be adjusted, component by component, in three spatial directions and/or spatial dimensions (in the example shown in FIG. **1** in the x-, y- and z-directions).

[0129] The beam splitting element **106** preferably takes the form of a 3D beam splitting element or comprises a 3D beam splitting element. The focus elements **120** may thus be formed, for example, such that they are each identical to one another and/or that they each constitute copies of one another.

[0130] With respect to the technical implementation and properties of the beam splitting element **106** designed as a 3D beam splitting element, reference is made to the scientific publication “Structured light for ultrafast laser micro- and nanoprocessing” by D. Flamm et al., arXiv:2012.10119v1 [physics.optics], Dec. 18, 2020. Express reference is made to the entire content thereof.

[0131] To carry out the beam splitting, in one embodiment of the beam splitting element **106**, in which the beam splitting element **106** is embodied, for example, as a 3D beam splitting element, a defined transverse phase distribution is imprinted on the transverse beam cross-section **112** of the input laser beam **108**. A transverse beam cross-section or a transverse phase distribution is to be understood in particular to mean a beam cross-section or a phase distribution in a plane orientated transverse and in particular perpendicular to the beam propagation direction **124** of the input laser beam **108**.

[0132] The focus elements **120** are formed by interference of the focused sub-beams **116**, and, for example, constructive interference, destructive interference or intermediate cases thereof can occur, such as partially constructive or partially destructive interference.
[0133] To form the focus elements **120** at the position x.sub.0, y.sub.0, z.sub.0 and/or with the distance d, the phase distribution imprinted by means of the beam splitting element **106** has a specific optical grating component and/or optical lens component for each focus element **120**.
[0134] Owing to the optical grating component, after focusing of the sub-beams **116**, a corresponding position offset of the focus elements **120** formed is produced in a first spatial direction and/or second spatial direction, for example in the x and/or y direction. Owing to the optical lens component, sub-beams **116** or sub-beam bundles impinge at different angles or with different convergence or divergence on the focusing optical unit **118**, which results after focusing in a position offset in a third spatial direction, for example in the z direction.
[0135] The intensity I of the respective focus elements **120** is determined by the phase positions of the focused sub-beams **116** relative to one another. These phase positions can be defined by the optical grating components and optical lens components mentioned above and can be selected in relation to one another when designing the beam splitting element **106** such that the focus elements **120** each have a desired intensity.
[0136] Alternatively or additionally, it is possible for the beam splitting element **106** to be formed as a polarization beam splitting element or to comprise a polarization beam splitting element. In this case, polarization beam splitting of the input laser beam **108** into beams which each have one of at least two different polarization states is carried out using the beam splitting element **106**.
[0137] In particular, the aforementioned polarization states should be understood to mean linear polarization states, and, for example, two different polarization states are provided and/or polarization states orientated perpendicular to one another are provided.
[0138] In particular, the polarization states are such that an electric field is orientated in a plane perpendicular to the beam propagation direction of the polarized beams (transverse electric).
[0139] For the polarization beam splitting, the beam splitting element **106** comprises, for example, a birefringent lens element and/or a birefringent wedge element. For example, the birefringent lens element and/or the birefringent wedge element are made of a quartz crystal or comprise a quartz crystal.
[0140] With regard to the mode of operation and design of the beam splitting element **106** as a polarization beam splitting element, reference is made to the German patent application with the reference number 10 2020 207 715.0 (filing date: Jun. 22, 2020), in the name of the same applicant, and to DE 10 2019 217 577 A1.
[0141] In particular, the sub-beams **116** can be formed with different polarization states by the polarization beam splitting. By focusing these sub-beams **116** by means of the focusing optical unit **118**, it is possible to form the focus elements **120** in each case from beams having a specific polarization state. The focus elements **120** can thus each be associated with a specific polarization state and/or implemented with a specific polarization state.
[0142] In particular, the focus elements **120** can be arranged and formed by polarization beam splitting by means of the beam splitting element **106** such that mutually adjacent focus elements **120** each have different polarization states.
[0143] For laser processing of the workpiece **104**, the focus elements **120** are introduced into the material **102** of the workpiece **104** and moved relative to the material **102** in the feed direction **126**, wherein the focus elements **120** are moved in particular at a specific feed rate in the feed direction **126**. In the example shown, the feed direction **126** corresponds to the y direction.
[0144] To carry out a relative movement of the focus elements **120** to the material **102**, the device **100** comprises a feed apparatus **127** (indicated in FIG. **1**). The feed apparatus **127** is configured to move the focus elements **120** in the feed direction **126** at a defined feed rate through the material **102**. For example, the feed apparatus can be realized by means of a workpiece holder which is

configured to move the workpiece **104** arranged thereon relative to the focus elements **120**.
[0145] The coupling-in of the focus elements **120**, which are introduced into the material **102** for laser processing of the workpiece **104**, takes place, for example, through a first outer side **130** of the workpiece **104**.
[0146] For example, the workpiece **104** is plate-like and/or panel-like and/or disc-like. A second outer side **132** of the workpiece **104** is arranged, for example, spaced apart in the thickness direction **134** and/or depth direction of the workpiece **104** from the first outer side **130**.
[0147] The material **102** of the workpiece **104** has, for example, an at least approximately constant thickness D with respect to the thickness direction **134**. The thickness D is, for example, 500 μm.
[0148] The feed direction **126** is orientated transversely of and in particular perpendicular to the beam propagation direction **124** and/or the thickness direction **134** of the workpiece **104**.
[0149] The formed focus elements **120** are preferably arranged such that they are positioned in a projection onto a projection plane **139** orientated transverse and in particular perpendicular to the feed direction **126** along a defined processing line **136** (cf. FIG. **2** and FIG. **4**). The aforementioned projection is understood to mean in particular an orthogonal projection of the focus elements **120** onto the projection plane.
[0150] The processing line **136** corresponds at least in portions to a target machining geometry with which the laser processing of the material **102** and in particular a subsequent severing of the material **102** is to be carried out.
[0151] The focus elements **120** are spaced apart from one another in the projection plane **139** and/or along the processing line **139** with an effective distance d.sub.eff. The effective distances d.sub.eff and intensities I of the focus elements **130** arranged along the processing line **136** are selected such that material modifications **138** are produced by applying the focus elements **120** to the material **102** and moving the focus elements **120** through the material **102** (FIG. **3**) and enable severing of the material along this processing line **136** and/or along a processing surface corresponding to this processing line **136**.
[0152] In particular, it is possible for the processing line **136** to extend between the first outer side **130** and the second outer side **132** and in particular continuously and/or without interruption between the first outer side **130** and the second outer side **132** of the workpiece **104**.
[0153] It is possible for the processing line **136** to have multiple different portions **140**. For example, in the example shown in FIG. **2**, the processing line **136** has a first portion **140***a*, a second portion **140***b* and a third portion **140***c*, wherein, with respect to the thickness direction **134**, the second portion **140***b* adjoins the first portion **140***a* and the third portion **140***c* adjoins the second portion **140***b*.
[0154] The processing line **136** is not necessarily designed to be regular and/or differentiable. For example, the processing line **136** can have irregularities. It is possible for the processing line **136** to have interruptions and/or gaps at which in particular no focus elements **120** are arranged.
[0155] The processing line **136** and/or different portions **140** of the processing line **136** can be formed, for example, as a straight line or curve.
[0156] Preferably, two or more planes **141** spaced apart from one another are provided, in each of which different sub-quantities of the formed focus elements **120** are arranged, and the planes **141** are spaced apart parallel to the feed direction **126**. For example, the planes **141** are each orientated parallel to the projection plane **139** and/or transverse and in particular perpendicular to the feed direction **126**.
[0157] In the example shown in FIGS. **5***a*, **5***b* and **6**, a sub-quantity of the formed focus elements **120** is positioned in a first plane **141***a* and a further plane, which in the exemplary embodiment shown is referred to as the second plane **141***b*. The first plane **141***a* and the second plane **141***b* are spaced apart from each other in the feed direction **126** and are orientated parallel to each other, for example. The focus elements **120** associated with the first plane **141***a* are hereinafter referred to as focus elements **120***a* and the focus elements **120** associated with the second plane **141***b* are

hereinafter referred to as focus elements **120***b*. In FIG. **4**, a total amount of the focus elements **120***a* and **120***b* is shown in the form of a projection of these focus elements **120***a* and **120***b* onto the projection plane **139**.
[0158] In particular, focus elements **120***a* and focus elements **120***b* are present, which are arranged in the projection plane **139** at different positions x.sub.0, z.sub.0.
[0159] A distance do between the first plane **141***a* and the second plane **141***b* is, for example, between 5 μm and 20 μm. Typically the distance do is approximately 10 μm.
[0160] The distance d already mentioned above is basically to be understood as the actual distance between adjacent focus elements **120** in the three spatial directions x, y, z and/or spatial dimensions.
[0161] The distance d between adjacent focus elements **120** is between 3 μm and 70 μm, preferably between 5 μm and 10 μm. In particular, the distance d between adjacent focus elements **120** which are arranged within a certain plane **141** lies in the mentioned ranges.
[0162] Gaps **143** are formed between adjacent focus elements **120** which are associated with a specific plane **141**, as shown in FIG. **5***a* using the example of the focus elements **120***a* of the first plane **141***a*.
[0163] The second plane **141***b* contains in particular focus elements **120***b*, which are positioned such that they lie in the gaps **143** when viewed in the projection plane **139**.
[0164] The respective effective distance d.sub.eff of the focus elements **120** provided for laser processing of the workpiece **104** can be selected differently for different focus elements **120** and/or different pairs of focus elements **120**. However, it is fundamentally also possible for the respective distance d to be at least approximately identical for all the focus elements **120** provided for laser processing of the workpiece **104**.
[0165] For example, it is possible for focus elements **120** having different effective distances d.sub.eff to each be associated with different portions **140** of the processing line. In particular, the distances d.sub.eff of the focus elements **120** associated with a specific portion **140** are then at least approximately constant.
[0166] In particular, an effective distance component d.sub.z,eff of the effective distance d.sub.eff orientated parallel to the thickness direction **134** of the material **102** and/or perpendicular to the feed direction **126** is different from zero for all focus elements **120** and/or for all pairs of mutually adjacent focus elements **120**. In particular, all adjacent focus elements **120** are spaced apart with a non-zero effective distance component d.sub.z,eff in the thickness direction **134**.
[0167] Furthermore, the processing line **136** and/or the portions **140** of the processing line **136** are associated with a specific angle of attack a and/or angle of attack range, which the processing line **136** or the portion **140** forms with the first outer side **130** of the workpiece **104**.
[0168] In the case of an angle of attack between 1° and 89°, the mutually adjacent focus elements **120** each have a further effective distance component d.sub.x,eff of the effective distance d.sub.eff which is different from zero and which is orientated perpendicular to the feed direction **126** and perpendicular to the effective distance component d.sub.z,eff.
[0169] The effective distance component d.sub.z,eff and the effective distance component d.sub.x,eff each lie in a plane orientated perpendicular to the feed direction **126**.
[0170] In the exemplary embodiment shown, the angle of attack a of the first portion **140***a* and the third portion **140***c* is 45° in magnitude and that of the second portion **140***b* is 90°.
[0171] By applying and/or introducing the focus elements **120** into the material **102**, localized material modifications **138** are produced in each case which are located at the local positions x.sub.0, y.sub.0, z.sub.0 of the corresponding focus elements **120** in the material **102** (FIG. **3**). By suitably selecting processing parameters, such as the distances d between the focus elements **120**, their intensities I, the feed rate orientated in the feed direction **126** and the laser parameters of the input laser beam **108**, the material modifications **146** can be produced, for example, as type III modifications, which are associated with a spontaneous formation of cracks **137** in the material **102**

of the workpiece **104**. In particular, cracks **137** are formed between mutually adjacent material modifications **146**.

[0172] Alternatively thereto, it is also possible by suitably selecting the processing parameters to produce the material modifications **146** as type I and/or type II modifications, which are accompanied by heat accumulation in the material **102** and/or by a change in a refractive index of the material **102**. The producing of the material modifications **146** as type I and/or type II modifications is associated with a heat accumulation in the material **102** of the workpiece **104**. In particular, to produce these material modifications **146**, the distance d between the focus elements **120** is selected so small that this heat accumulation occurs when the focus elements are applied to material **102**.

[0173] FIG. **7***a* shows a simulated intensity distribution of a plurality of focus elements **120**, wherein the distance d is approximately 17.5 μm for these focus elements **120**. In the grayscale value representation shown, lighter regions represent higher intensities.

[0174] FIG. **7***b* shows a simulated intensity distribution of a plurality of focus elements **120**, wherein the distance d is approximately 8.0 μm.

[0175] The laser processing of the workpiece **104** by means of the device **100** functions as follows:

[0176] To carry out the laser processing, the focus elements **120** are applied to the material **102** of the workpiece **104** and the focus elements **120** are moved in the feed direction **126** relative to the workpiece **104** through its material **102**.

[0177] In this case, the material **102** is a material transparent to a wavelength of laser beams from which the focus elements **120** are each formed, such as a glass material. In the example shown, the focus elements are formed by beam forming of the input laser beam **108**.

[0178] By applying the focus elements **120** to the material **102**, material modifications **138** are produced in the material **102** which are arranged in a cross-section orientated perpendicular to the feed direction **126** along the processing line **136** (FIG. **8***a*). In the example shown in FIG. **8***a*, material modifications **138** are produced continuously over the entire thickness D of the material **102**.

[0179] By relative movement of the focus elements **120** in relation to the material **102** along the trajectory **142**, a processing surface **144** corresponding to the processing line **136** is formed, on which the material modifications **138** are arranged. This results in a planar formation and/or arrangement of the material modifications **146** along the processing surface **144**.

[0180] The trajectory **142** can in principle have straight and curved portions. In the case of curved portions, the processing line **136** is in particular turned during the laser processing so that it always lies in a plane orientated perpendicular to the feed direction **126**. This can be realized, for example, by corresponding rotation of the beam splitting element **106** or by relative rotation of the entire device **100** in relation to the workpiece **104**.

[0181] A distance between adjacent material modifications **138** in the feed direction **126** can be defined, for example, by adjusting a pulse duration of the input laser beam **108** and/or by adjusting the feed rate.

[0182] The material modifications **146** produced along the processing line **136** result in particular in a reduction in a strength of the material **102**. This allows the material **102** to be severed into two different workpiece segments **146***a*, **146***b* after the material modifications **146** have been produced on the processing surface **144**, for example by exerting a mechanical force (FIG. **8***b*).

[0183] The workpiece segment **146***a* in the example shown is a yield segment having a severing surface **148**, which has a shape corresponding to the shape of the processing line **136**. In this case, the workpiece segment **154***a* is a residual workpiece segment and/or scrap segment.

[0184] For example, the material **102** of the workpiece **104** is quartz glass. For example, to produce the material modifications **138** as type-I and/or type II modifications, a laser beam from which the focus elements **120** are formed has a wavelength of 1030 nm and a pulse duration of 1 ps. Furthermore, a numerical aperture associated with the focusing optical unit **118** is then 0.4 and a

pulse energy associated with a single focus element **120** is then 50 to 200 nJ.
[0185] To produce the material modifications **138** as type III modifications, the pulse energy associated with a single focus element **120** is 500 to 2000 nJ, with otherwise identical parameters.
[0186] While subject matter of the present disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. Any statement made herein characterizing the invention is also to be considered illustrative or exemplary and not restrictive as the invention is defined by the claims. It will be understood that changes and modifications may be made, by those of ordinary skill in the art, within the scope of the following claims, which may include any combination of features from different embodiments described above.
[0187] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.
LIST OF REFERENCE SIGNS
[0188] α Angle of attack [0189] d Distance [0190] d.sub.0 Distance [0191] d.sub.eff Effective distance [0192] d.sub.x,eff Effective distance component [0193] d.sub.z,eff Effective distance component [0194] D Thickness [0195] I Intensity [0196] x.sub.0 Position in the x-direction [0197] y.sub.0 Position in the y-direction [0198] z.sub.0 Position in the z-direction [0199] **100** Device [0200] **102** Material [0201] **104** Workpiece [0202] **106** Beam splitting element [0203] **108** Input laser beam [0204] **110** Laser beam source [0205] **112** Beam cross-section [0206] **114** Wavefront [0207] **116** Sub-beams [0208] **116***a* Sub-beam [0209] **116***b* Sub-beam [0210] **118** Focusing optical unit [0211] **120** Focus element [0212] **120***a, b* Focus element [0213] **121** Focus distribution [0214] **122** Beam forming apparatus [0215] **124** Beam propagation direction [0216] **126** Feed direction [0217] **127** Feed apparatus [0218] **130** First outer side [0219] **132** Second outer side [0220] **134** Thickness direction [0221] **136** Processing line [0222] **137** Crack [0223] **138** Material modification [0224] **139** Projection plane [0225] **140** Portion [0226] **140***a* First portion [0227] **140***b* Second portion [0228] **140***c* Third portion [0229] **141** Plane [0230] **141***a* First plane [0231] **141***b* Second plane [0232] **142** Trajectory [0233] **143** Gap [0234] **144** Processing surface [0235] **146***a* Workpiece segment [0236] **146***b* Workpiece segment [0237] **148** Severing surface

Claims

1. A method for laser processing of a workpiece which has a transparent material, the method comprising: dividing an input laser beam into a plurality of sub-beams by using a beam splitting element, focusing the sub-beams coupled out of the beam splitting element, wherein multiple focus elements are formed by the focusing the sub-beams, and applying the focus elements to the material of the workpiece for the laser processing, wherein the focus elements are moved relative to the material in a feed direction, and wherein a first sub-quantity of the focus elements is arranged in a first plane, and a second sub-quantity of the focus elements is arranged in at least one other plane, the first plane and the at least one other plane being spaced apart in the feed direction, and the first plane and the at least one other plane being orientated perpendicular to the feed direction.

2. The method according to claim 1, wherein the focus elements are positioned such that, when the focus elements are viewed in a projection plane orientated perpendicular to the feed direction, at least a sub-quantity of the focus elements associated with the first plane and the focus elements associated with the at least one other plane is arranged at different spatial positions in the projection plane.

3. The method according to claim 1, wherein the focus elements are positioned such that, when the focus elements are viewed in a projection plane orientated perpendicular to the feed direction, at least a sub-quantity of the focus elements associated with the first plane and the focus elements associated with the at least one other plane is positioned in the projection plane along a processing line, wherein, by the movement of the focus elements relative to the material in the feed direction, material modifications are produced in the material along the processing line.

4. The method according to claim 3, wherein, by the movement of the focus elements relative to the material, the material modifications are produced along a processing surface corresponding to the processing line, wherein the material modifications are arranged in a cross-section, orientated perpendicular to the feed direction, through the processing surface along the processing line.

5. The method according to claim 3, wherein adjacent focus elements arranged along the processing line have an effective distance of at least 2 μm and/or at most 200 μm in the projection plane.

6. The method according to claim 3, wherein an effective distance of adjacent focus elements arranged along the processing line in the projection plane and/or an intensity of the focus elements is selected such that, by the applying the focus elements to the material and movement of the focus elements relative to the material in the feed direction, the material modifications are formed in the material which enable a severing of the material.

7. The method according to claim 3, wherein at least a sub-quantity of adjacent focus elements arranged along the processing line is at least approximately spaced apart in the projection plane with a same effective distance from one another.

8. The method according to claim 3, wherein at least a sub-quantity of adjacent focus elements arranged along the processing line is spaced apart at an effective distance, wherein the effective distance has an effective distance component different from zero that is orientated parallel to a thickness direction of the workpiece, and/or has a further effective distance component different from zero that is orientated perpendicular to the thickness direction of the workpiece.

9. The method according to claim 3, wherein an angle of attack between the processing line and an outer side of the workpiece, through which the focus elements for the laser processing are coupled into the material of the workpiece, is at least 1° and/or at most 90°, at least in some portions.

10. The method according to claim 1, wherein the focus elements are positioned such that, when the focus elements are viewed in a projection plane orientated perpendicular to the feed direction, gaps are formed between adjacent focus elements that are associated with the first plane, wherein the second sub-quantity of focus elements associated with the at least one further plane are arranged in the gaps.

11. The method according to claim 1, wherein a first distance between adjacent focus elements that are associated with the first plane is at least 3 μm and/or at most 70 μm and, and/or a second distance between adjacent focus elements that are associated with the at least one other plane is at least 3 μm and/or at most 70 μm.

12. The method according to claim 1, wherein the dividing the input laser beam by using the beam splitting element comprises phase imprinting on a beam cross-section of the input laser beam.

13. The method according to claim 1, wherein material modifications are produced in the material by the applying the focus elements to the material of the workpiece, wherein the material modifications are accompanied by a crack formation of the material, and/or wherein the material modifications are type III material modifications.

14. The method according to claim 1, wherein material modifications are produced in the material by the applying the focus elements to the material of the workpiece, wherein the material modifications are accompanied by a change in a refractive index of the material, and/or the material modifications are type I material modifications and/or type II material modifications.

15. A device for laser processing of a workpiece having a transparent material, the device comprising: a beam splitting element for dividing an input laser beam into a plurality of sub-beams, a focusing optical unit for focusing the sub-beams coupled out of the beam splitting element, wherein multiple focus elements for the laser processing of the workpiece are formed by the focusing the sub-beams, and a feed apparatus for moving the focus elements relative to the material of the workpiece in a feed direction, wherein the beam splitting element and the focusing optical unit are configured to arrange the focus elements in such a way that a first sub-quantity of the focus elements is arranged in a first plane, and a second sub-quantity of the focus elements is arranged in at least one other plane, the first plane and the at least one other plane being spaced apart in the feed direction, and the first plane and the at least one other plane being orientated perpendicular to the feed direction.